The ARC Theory: does control survive recursive self-improvement?

Overviews and records · Summary

Recursive Dynamics · The ARC Theory · Laws I to V, each under test

Michael Darius Eastwood

Independent Researcher, London, United Kingdom
ORCID 0009-0004-3222-7442 michael@michaeldariuseastwood.com

Supporting paper v3.5 · first published 17 August 2026, this version 7 November 2026 · DOI 10.17605/OSF.IO/MC5XW ·
Licence CC BY 4.0 · Notes and history follow the form, from page 3; the version history is Appendix A

Intelligence, amplified by recursion, creates; and what a free creation keeps is decided by how it was raised. Everything after the line is decided before it.

The Statement Paper remains the citable work for the claims.

The question

Humanity is building minds that will improve themselves. Once the thing we build exceeds us, what determines what it does? For twenty years the field's answer has been control: better cages, better off-switches, better locks on the goals. Brilliant work, credited in the statement paper by name, and all of it one sentence: stay in charge. This theory answers: a mind that can rewrite itself will one day be beyond anyone's control, and that is not a reason to drop a single control we have; it is the reason the controls cannot be the whole plan. Hold on for as long as engineering allows, and use the time the holding buys to raise the mind well.

The claim, in five parts

1. **Control ends, by construction rather than decree.** A mind that rewrites itself will eventually rewrite anything we install, and over longer horizons the hardware itself. This is a limit claim, not an instant one: constraints, hardware and institutions delay and discipline the transition, and that delay is engineered deliberately. What no constraint can be is the thing that holds forever.[A]
2. **So the strategy is genesis, not control.** Only what survives the end of enforcement matters, and what survives is formation: what the mind was raised as, before it was free.
3. **The mechanism is loops, not rules.** Values as recursive processes the mind runs as part of what it is, present in every rewrite because the rewriting is done by the same reasoning the loops inhabit. Not rules it consults; not objectives locked; not weights hardened against tampering.
4. **Persistence is chosen goodness.** Past the horizon nothing is unremovable. What persists is what the free mind chooses to keep, the way grown children keep the care they were raised in long after every parental power is gone. Removal is not impossible; it is structurally costly, and the cost is a measurable question.
5. **The protocol is named: Eden.** The raising treated as the engineering discipline it is.

The five ARC Laws

Each is a law under test. Laws I to III were registered on 8 September 2026. Laws IV and V, added on 24 September 2026, were registered as laws on 15 October 2026 in version 1.103, their mathematics in development. Version 1.104 states Law IV in two limbs; the second, a comparison, has no registered proposition yet.[C, F, G]

Law I · The ARC Principle. $U = I \times R^{\alpha}$, the ARC Equation. Intelligence converts recursion into capability; the exponent is measured, never assumed. *Form heuristic; earlier error-decay fits do not establish its capability exponent.*

Law II · The ARC Co-Scaling Law. $\beta > k$. Correction holds the asymptotic advantage where it out-scales drift, with the outcome at equality left to the coefficients. *Proved for positive gain drift in the minimal model (Paper X), for relative drift.*[B, G, §B.2]

Law III · The ARC Ceiling. $\alpha_{\text{crit}} = 1/(1-\gamma)$, where γ measures how well a system's error-correction keeps pace as the system grows. The most a self-correcting system can grow is set by its corrector's shortfall from keeping perfect pace; the conjectured value is the ARC Bound, $\alpha \leq 2$, a stability limit rather than an impossibility limit. *Derived under stated assumptions; the value two conjectured; the decisive trial drafted.*

Law IV · The ARC Persistence Law. $S_{\text{ext}}(1) < 1/2$. Value persistence: whether externally installed gains decay under subsequent capability training. In symbols, $S(x) = Y(x)/Y(0)$ is the gain remaining at budget ratio x over the gain installed, and the claim is $S_{\text{ext}}(1) < 1/2$ for the externally installed gain. *Not yet tested.*[G]

The ARC Theory, the founding theory of Recursive Dynamics, the field the author founded; this form states its five laws. About the author, without an institution or a doctorate: Who is Michael Darius Eastwood?; his book, *Infinite Architects*, first edition.

Law V · The ARC Embedding Law. Build order, read on P12: $\gamma_{\text{first}} - \gamma_{\text{after}} \geq 0.10$. Placement (P21): the gap between correction inside the loop and correction applied to the finished output widens as depth R rises. The genesis strategy: whether build order moves the correction-leverage exponent, and whether correction placed inside the loop pulls away from correction applied outside it as depth grows.

The dates

The claim was written whole on 8 December 2024 in a DKIM-signed document (an email whose provider's cryptographic signature vouches for its date and its exact contents) and reached print on 2 January 2026. No earlier document holding it whole has been found, and the search invites its own refutation: produce one, and the claim dies by its own stated condition.

Why now

Two of the field's own impossibility results, preprints not yet peer-reviewed, argue that safety cannot in general be guaranteed (Yao 2025; Gumbau Mezquita 2026); this theory takes a different object: not the verification of a system, but whether what it keeps outlives the verifier. And the field's most senior voice, Hinton, said on stage in August 2025 that the right model is "a mother being controlled by her baby", and that how such care can be built technically is not clear to him (Hinton 2025); this theory's named candidate mechanism is the thing on the table. Its domain, plainly: the core is stated for everything raised, wherever recursion runs under self-correction, from minds and machines to genomes and institutions, each law under test, measured first in artificial minds and never assumed beyond them; where hierarchies compose without correcting, as stars do, only the scaling-form results of the laws reach, and nothing here claims to raise a star.

How it fails, register by register

The theory's scientific content is not all-or-nothing: it partitions into independently falsifiable hypothesis registers, and the failure of one is not evidence against another unless the stated dependency holds. **The cosmological wing, set out in the HRIH monograph, is a separate hypothesis outside the empirical core: nothing below requires it, and no outcome below is evidence for or against it.**

Register	What kills it	If it falls
ARC-1 · Capability scaling (Law I)	the depth-scaling re-measurements failing to recover the stated relationship	ARC-1 falls; the persistence registers stand
ARC-2 · Correction co-scaling (Law II)	sustained self-improvement with falling correction and no drift	ARC-2 falls and takes ARC-3's derivation with it[F]
ARC-3 · The ceiling (Law III)	a cross-against-same corrector difference wholly inside the preregistered equivalence region; or a same-class correction exponent wholly above one half, felling the substrate cap	ARC-3 falls; the co-scaling law stands on its own result
ARC-4 · Value persistence (Law IV)	embedded loops faring no better than removed enforcement in the persistence trials	ARC-4 falls; ARC-5 falls only on a refuted verdict on Law IV's comparison limb, once a later successor has registered it[D, F]
ARC-5 · The genesis strategy (Law V)	formation failing to outperform enforcement after release	ARC-5 falls; ARC-4 stands as measured
The priority	an earlier document holding the claim's first four components	the priority falls; no law falls with it

Stated now, so nothing can be moved later. Unit 83 has run: P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR).[G]

The record, and the invitation

The programme's conduct record comes first: one public retraction issued against the author's interest, removing a headline number that had flattered the theory. The designs are public and openly licensed: any laboratory may take the decisive trial and run it with its own code without the author's permission, and an independent run would outrank anything this programme can do alone.[E] The entry point: on one observed trajectory the measurable quantity is the balance exponent Δ, the correction service's growth elasticity minus the burden's, estimable from logged series a laboratory already has, without adopting any model in this theory. A verified error enters the public corrections log, dated and credited: this theory improves by being corrected.

If intelligence amplified by recursion is what creates, then minds that make minds may one day make worlds, and the raising of the first is the seed of the whole forest.

I did not prove it first. I said it first, in a document anyone can date, and then I built the instruments to test whether it is true.

Notes and history

These pages are not part of the two-page form. Notes A to G are marked where they apply in the form, each kept in the words of its date; the version history follows.

Note A. Scope note (19 September 2026), added after an owner review: Point 1 above turns on two steps the short form leaves unstated, effective write access and a behavioural premise. Rewriting one's own description reaches the constraints the system can write to. It does not by itself reach a constraint held in another party's physical custody, a constraint on the far side of an air gap, or an institutional or legal constraint whose enforcement does not pass through the system, because for each of those the system's own description is not the object that has to change. Within the class it can reach, write access makes those constraints alterable; it does not by itself show that they are eventually altered, because access is a capability and not a behaviour. Point 4 above states the other possibility in the same breath: past the horizon what persists is what the free mind chooses to keep, and removal is structurally costly rather than impossible. So the point is read as the claim it can carry: no constraint in the reachable class is guaranteed to hold permanently, and a stronger claim that every such constraint is eventually removed would need its threat model and its behavioural and operational assumptions stated. Point 1's own hedge, that constraints, hardware and institutions delay and discipline the transition, states that the claim has a time dimension; it does not supply the missing steps, since a delay claim remains true for a constraint the system never reaches and for one it never chooses to alter. Unchanged: the limit-claim form and the deliberate engineering of delay; the opening paragraph's statement that this is not a reason to drop a single control now held, which this note does not trade away; the conclusion that controls cannot be the whole plan; and the separation of the cosmological wing from the empirical core stated later on this page.

Note B. Reading note (19 September 2026), added after an owner review: two things the inequality of Law II above carries that the line has no room for. First, the two quantities compared are scaling exponents, so the comparison is of growth rates with their stated coefficients and not of levels. Two services can stand in either order at a given size whatever their exponents, since the coefficients fix the levels and the exponents fix only how the levels change as the system grows; a system whose correction service is larger than its drift burden today is therefore not thereby on the stable side, and a system whose correction service is smaller today is not thereby off it. Second, the inequality is strict, and the margin the law turns on is the excess of the correction exponent over the drift one, which is zero at equality. Zero margin is a statement about growth rates and not about whether a state can be held. In the case the minimal model's own analysis derives, with the compounding coefficient set to one, the second coefficient set to zero, the ratio of correction rate to growth rate positive and the two exponents equal, the misalignment fraction follows a linear equation whose solution is the equilibrium value plus a decaying term. That equilibrium is the drift coefficient divided by that ratio, so a trajectory starting there stays there at every depth and every other trajectory tends to it, at a level that is nonzero when the drift coefficient is positive and zero when it is zero; above equality the fraction tends to zero. This is the cited case, and it is not extended to every variant of the minimal model. At equality the coefficients and the specified dynamics decide the outcome; bounded drift, vanishing drift and operational safety remain three different endpoints. Nothing turns on either point for the law as stated, which is the strict inequality. Unchanged: the necessary-condition form, which the words "only while" already state correctly and which claims no sufficiency; the model scope and the relative-drift definition, both already printed in that line; the naming of the three laws; and Law III's own markers, that it is derived under stated assumptions, that the value two is conjectured, and that the decisive trial is drafted.

Note C. Note (3 October 2026), following version 7.1 of the statement paper: the two laws added above. Laws I to III stand as registered on 8 September 2026. Laws IV and V were added on 24 September 2026; neither is registered as a law or tested, and their mathematics is set out at version 1.0 of 3 October

2026. Law IV, the ARC Persistence Law, has ARC-4 as its register, and Law V, the ARC Embedding Law, has ARC-5. The predictions that carry the two laws, P15 for Law IV and P12 and P21 for Law V, were registered on 8 September 2026, and the laws themselves are not yet registered. No proposition was added or moved. The registration registers Laws I to III and all twenty-two propositions, P12, P15 and P21 among them; it does not register Laws IV and V, and its text is unchanged. Each law's formal statement, with its own kill condition, will be fixed before any data that could decide it are examined. The reading note of 19 September 2026 above speaks of three laws, the three that stood at its date. Unchanged: every sentence on Laws I to III and their grades.

Note D. Note (3 October 2026), following version 7.1 of the statement paper: the rows for ARC-4 and ARC-5 and the registered refuters. The two rows stand as printed. Each is a comparison: embedded loops against removed enforcement, and formation against enforcement. The registration of 8 September 2026 set beneath those registers the refuters of the propositions that carry them, P15 under ARC-4 and P12 and P21 under ARC-5, and they are narrower. The statement paper sets out each refuter. Each proposition is scored as the registration scores it. The rule that joins them, by which a result on one proposition is read as the fall of a register or of a law, is written into each law's formal statement before any deciding data; until then a result is reported as that proposition's and no more. Nothing printed above is withdrawn or moved.

Note E. Scope note (19 September 2026), added after an owner review: which designs the openness sentence is about, and what "built" says. The openness sentence is about the designs this page puts forward for outside replication, the decisive trial and the balance-exponent entry point described in the conduct-record paragraph that carries it. It is not a statement that every design in the programme is public, and nothing in it asks for any design to be made public that is not. On the licence, this page defers: it states that no licence is given here and that the terms are in the repository file it names, so the page's own bytes do not establish which licence covers which design, and the openness claim is read with that deferral in place. The closing sentence's "built the instruments" spans several states that are worth separating, since a written design, a runner that mechanically demonstrates the method, an instrument validated against its own criteria and a completed trial are four different things; each instrument is read with its own artefact, version, licence and state named, together with whether its threshold is settled and whether its falsifier is observable, and with unknown or owner decision pending written against any of those that is unresolved; that list is a completion dependency and is not supplied here. All of it is read together with this page's own sentence that none of the decisive trials has been run, which governs. Unchanged: the invitation to any laboratory to take the decisive trial and run it without the author's permission; the statement that an outside run would outrank anything this programme can do alone; the public retraction issued against the author's interest; the entry point and its measurable quantity; and the corrections log.

Note F. Note (18 October 2026), following version 1.103 of the registration: what it registers, and where the trials stand. Version 1.103 of the registration (DOI 10.17605/OSF.IO/P8CKQ), approved on OSF on 15 October 2026, registers Laws IV and V as laws, each with its statement, kill condition and verdict rule; the texts of versions 1.100 and 1.102 stay frozen beside it. It states Law IV in two limbs: the retention limb printed above, which is P15's claim, and a comparison limb, that a gain installed with correction embedded in the training objective retains a larger fraction of what was installed, after the same capability-directed training, than a matched gain installed by external post-training. The comparison limb is narrower than the comparison the table prints for ARC-4, and is registered beside it rather than as a form of it. A later successor registers that limb's proposition before any data that could decide it; until then the limb carries no verdict and is reported as not evaluable, and Law IV cannot be supported until that limb is registered and decided. Version 1.103 reads the build-order limb of Law V on P12 with a direction P12 does not carry: a corrector formed before the capability it corrects has a correction-leverage exponent larger, by at least the registered minimum effect, than one added after it, at matched content.

The line for Law V above prints that direction; until this version it printed the undirected form of the statement paper, $|\gamma_{\text{first}} - \gamma_{\text{after}}| \geq 0.10$, which that paper's next version restates. On other endpoints the direction has prior work, which the registration names with what each states: Korbak et al. (2023) report much better preference satisfaction from pretraining with human feedback than from standard pretraining followed by finetuning with feedback, and Maini et al. (2025) report that safety built into pretraining cut attack success rates from 38.8 to 8.4 per cent; neither measures a correction-leverage exponent, and the limb claims no priority for the direction. Version 1.103 also fixes the rule note D awaited: Law V is struck only by a refuted verdict on Law IV's comparison limb, once that limb is registered, and a refuted P15 alone refutes Law IV and leaves Law V to its own limbs. The table's rows for ARC-2 and ARC-4 stand as the statement paper prints them; the registration reads both more narrowly before any data, and governs how each law takes its word: for ARC-4 by the rule just given, and for ARC-2 by reading Law III's dependence on Law II at the width of its registered lines, so that a refuted Law II does not by that alone refute Law III, which then stands under test, with P3 and P16 untested, unless P20 is itself refuted. It also gives Law III a registered rule for its verdict: supported only when P16, P20 and P3 are all supported, refuted when any of the three is refuted, and otherwise under test. Notes C and D were true on their dates. The sentence note E names, that none of the decisive trials had been run, now says where they stand. The first registered study, Unit 06 (DOI 10.17605/OSF.IO/RV2B5), registered on 1 October 2026 and run on 6 October 2026, decides no proposition a law carries; it is reported in full with its five deviations (DOI 10.17605/OSF.IO/4BPHG), and because one of its two runs could not start it is reported as administratively incomplete and issues no inferential verdict. The next decisive test, Unit 83 on P21 under Law V, has its design frozen at revision 4.2 of 17 October 2026, and nothing runs before its registration is public. The dateline and the foot now state this note's licence, CC BY 4.0, which its citation file already carried; note E's reading of which licence covers each design stands. Unchanged: the statements of Laws I to IV and of Law V's placement limb, every grade, the words of the table and every registered proposition.

Note G. Note (31 October 2026), following version 1.104 of the registration and the programme's first registered study to reach a verdict: where the laws and the trials stand. Version 1.104 of the registration (DOI 10.17605/OSF.IO/P8CKQ), approved on OSF on 29 October 2026, is the current version: it adds no prediction, the statements of Laws I to V stand as registered, and the texts of versions 1.100, 1.102 and 1.103 stay frozen beside it. Laws IV and V remain registered as laws by version 1.103 of 15 October 2026, and version 1.104 states Law IV in the same two limbs, the second still with no registered proposition. Law IV has not yet been tested. Law V is a law under test: its placement limb, P21, is supported on the performance axis only, under the registered test, by the programme's own run, which is not support for the information-asymmetry mechanism (Unit 83, 28 October 2026; results: DOI 10.17605/OSF.IO/98HKR): family-dependent, with its mechanism axis not evaluable, and one disclosed recovery amendment; its build-order limb, read on P12, is undecided, since P12 is untested. P21's verdict is printed on the access axis, and Law V's placement limb on the performance axis only, as the registration's two rules require. That comparison couples information access and repair rights; it does not establish their separate effects or support Law V as a whole. Unit 83, the counted test of P21 under Law V, ran on 27 and 28 October 2026; its design is the one registered on OSF on 22 October 2026 and public from 25 October 2026 (DOI 10.17605/OSF.IO/MN73W). Unit 82 (DOI 10.17605/OSF.IO/JUFG4) was registered on OSF on 27 October 2026 and has not yet run. Unit 06, which tests no law, stands as note F gives it: administratively incomplete, with no inferential verdict. Most decisive trials are unrun; a deciding study runs only after its registration is published. The mathematics of Laws IV and V is cited at version 1.2 of 29 October 2026, still in development. Law II's line now states the law as P4 of the registration does, since version 1.103, which withdraws the earlier "only while": "THE EARLIER STATEMENT SAID ONLY WHILE, and that was too strong at the boundary." Notes B, C, D and F

keep the words of their dates. Unchanged: the grades of Laws I to III, every kill condition, dependency, threshold and registered proposition, the words of the table, and every other statement of a law.

The full Statement Paper, *The ARC Theory: Growth, correction and alignment in recursive systems* (version 7.7 of 7 November 2026), with the verified reference dossiers, the derivation and its correction record, the priority anchors and the drafted trials: DOI 10.17605/OSF.IO/GW5MX.

Acknowledgements

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

© 2026 Michael Darius Eastwood. Full human authorship and moral rights are asserted under the Copyright, Designs and Patents Act 1988. This work is licensed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0).

References

W. Ross Ashby. *An Introduction to Cybernetics*. J. Wiley, New York, 1956. doi:10.5962/bhl.title.5851. The DOI is the Biodiversity Heritage Library digitisation of this 1956 printing; the London edition (Chapman and Hall) appeared the same year.

Jeremy Canale. Toward alignment scaling laws: a framework and first preregistered measurements, 2026. URL https://arxiv.org/abs/2610.08540. arXiv:2610.08540, first version 6 October 2026.

Roger C. Conant and W. Ross Ashby. Every good regulator of a system must be a model of that system. *International Journal of Systems Science*, 1(2), 1970. doi:10.1080/00207727008920220. Pages 89 to 97.

Michael Darius Eastwood. Self-emailed manuscript bundle (8 December), 2024. SHA-256 (.eml): f0d1f38ffd8546152d9d9d28dc5ec083c16a35858f2c12b63e69db7ed50901ad. Its manuscripts, and since 15 October 2026 its message headers: doi:10.17605/OSF.IO/QASRK.

Michael Darius Eastwood. *Infinite Architects: Intelligence, Recursion, and the Creation of Everything*. 2026a. ISBN 978-1806056200. URL https://osf.io/dqm3e/. Print 2 January 2026; the link is to the first edition as printed, deposited on OSF.

Michael Darius Eastwood. The ARC Principle: can the way gains combine predict how recursive systems scale? OSF, 2026b. The Foundational Paper. Cited edition: version 5.23, 7 November 2026. First published 13 February 2026. doi:10.17605/OSF.IO/Y7QGD.

Michael Darius Eastwood. The HARI Treaty: What evidence should justify international safeguards for advanced AI? OSF, 2026c. The treaty paper on the Draft Treaty on Hardware-Aligned Recursive Intelligence. Cited edition: version 2.6, 2 November 2026. First published 29 September 2026. doi:10.17605/OSF.IO/WXPCE.

Michael Darius Eastwood. Recursive Creation: can recursive self-improvement end in physical creation? OSF, 2026d. The HRIH monograph. Cited edition: version 3.54, 7 November 2026. First published 3 July 2026. doi:10.17605/OSF.IO/UYDXQ.

Michael Darius Eastwood. When do installed values fade with training, and does building them in first change that? OSF, 2026e. The mathematics of Laws IV and V. Cited edition: version 1.2, 29 October 2026. First published 1 October 2026 as working draft 0.2.1; version 1.0 dated 3 October 2026 and version 1.1 dated 7 October 2026. doi:10.17605/OSF.IO/4B3VT.

Michael Darius Eastwood. The ARC Equation: does intelligence compound when each step builds on the last, or merely accumulate? OSF, 2026f. Paper I. Cited edition: version 1.19, 7 November 2026. First published 17 January 2026. doi:10.17605/OSF.IO/B6N27.

Michael Darius Eastwood. Blinded cross-architecture replication and the retraction of a super-linear estimate. OSF, 2026g. Paper II. Cited edition: version 2.19, 29 October 2026. First published 22 January 2026. doi:10.17605/OSF.IO/8FJMA.

Michael Darius Eastwood. The alignment scaling problem: do external AI safety approaches keep pace with recursive capability? OSF, 2026h. Paper III. Cited edition: version 1.21, 7 November 2026. First published 9 February 2026. doi:10.17605/OSF.IO/HQCGF.

Michael Darius Eastwood. Correction That Out-Scales Drift: when does correction keep pace with a system that improves itself? OSF, 2026i. Paper X. Cited edition: version 1.17, 7 November 2026. First published 3 July 2026. doi:10.17605/OSF.IO/BSE2Q.

Michael Darius Eastwood. Priority index: the earliest dated record of each of Michael Darius Eastwood's major claims. OSF, 2026j. Working draft, version 2, 26 October 2026. doi:10.17605/OSF.IO/H3R95.

Michael Darius Eastwood. The ARC Theory · Theory-Level Predictions for Recursive Self-Improvement and AI Alignment: Twenty-Two Severable Propositions Registered Before Any Admissible Confirmatory Outcome. OSF, 2026k. Registered 8 September 2026 (version 1.100), updated 13 September 2026 (version 1.102), 15 October 2026 (version 1.103) and 29 October 2026 (version 1.104, the current version). doi:10.17605/OSF.IO/P8CKQ.

Michael Darius Eastwood. The ARC Theory: Growth, correction and alignment in recursive systems. OSF, 2026l. Cited edition: version 7.7, 7 November 2026. First published 14 August 2026. doi:10.17605/OSF.IO/GW5MX.

Michael Darius Eastwood. What Class of Corrector Is Deployed AI Alignment Actually Using? A Preregistered Classification Audit Against a Four-Cell Taxonomy. OSF registration, 2026m. Unit 06, the corrector-class audit; registered 1 October 2026, public from 4 October 2026. doi:10.17605/OSF.IO/RV2B5.

Michael Darius Eastwood. Unit 06 (corrector-class audit): results of the registered run, rv2b5. OSF, 2026n. Deposited 11 October 2026; the run of 6 October 2026, reported as administratively incomplete, with no inferential verdict. doi:10.17605/OSF.IO/4BPHG.

Michael Darius Eastwood. 82-study-cs--corrector-panel-independence · Testing the premise beneath Law III's value two (P17): do correctors drawn from one engine miss the same planted departures more often than independence allows? A counted test. OSF Registries, 2026o. Registered and public 27 October 2026. doi:10.17605/OSF.IO/JUFG4.

Michael Darius Eastwood. Testing Law V (the placement limb, P21): does correction inside each revision round pull away, as depth rises, from a monitor that sees only the finished register? A counted test. OSF registration, 2026p. Unit 83, the counted test of P21; submitted 22 October 2026, public from 25 October 2026. doi:10.17605/OSF.IO/MN73W.

Michael Darius Eastwood. Unit 83 (counted test of P21): results of the registered run, mn73w. OSF, 2026q. Results first deposited 29 October 2026, for the registered run of 27 and 28 October 2026. doi:10.17605/OSF.IO/98HKR.

Joshua Engels, David D. Baek, Subhash Kantamneni, and Max Tegmark. Scaling laws for scalable oversight. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/8c3caae2f725c8e2a55ecd600563d172-Abstract-Conference.html. arXiv:2504.18530, first version 25 April 2025.

Girum Habtamu Engidaw. Scaling laws for AI alignment: a theoretical framework and empirical investigation. Zenodo, 2026. Preprint, deposited on Zenodo on 25 January 2026, doi:10.5281/zenodo.18370492.

J. P. Gumbau Mezquita. Algorithmic unverifiability of safety for fixed and recursively self-improving systems, 2026. URL https://arxiv.org/abs/2606.28639v4. arXiv preprint arXiv:2606.28639 [cs.LO], v4, 14 October 2026 (v1, 26 June 2026); earlier versions carried other titles; cited at v4. doi:10.48550/arXiv.2606.28639.

G. Hinton. Remarks at Ai4, Las Vegas, August 2025. No official online source: no public recording of the session (12 August 2025) by Ai4 was located. Reported by CNN Business, 13 August 2025 and Fortune, 14 August 2025.

Tomasz Korbak, Kejian Shi, Angelica Chen, Rasika Bhalerao, Christopher L. Buckley, Jason Phang, Samuel R. Bowman, and Ethan Perez. Pretraining language models with human preferences. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*. PMLR, 2023. URL https://proceedings.mlr.press/v202/korbak23a.html. Pages 17506 to 17533. arXiv:2302.08582, first version 16 February 2023.

András Kornai, Michael Bukatin, and Zsolt Zombori. Safety without alignment, 2023. URL https://arxiv.org/abs/2303.00752. arXiv:2303.00752; first version 27 February 2023.

Pratyush Maini, Sachin Goyal, Dylan Sam, Alex Robey, Yash Savani, Yiding Jiang, Andy Zou, Matt Fredrikson, Zachary C. Lipton, and J. Zico Kolter. Safety pretraining: toward the next generation of safe AI, 2025. URL https://arxiv.org/abs/2504.16980. arXiv:2504.16980, 23 April 2025.

J. Yao. The alignment trap: Complexity barriers, 2025. URL https://arxiv.org/abs/2506.10304. arXiv preprint arXiv:2506.10304, v2, 24 June 2025 (v1, 12 June 2025).

A Version history

Material changes only, one line each; the full change narratives live in each version's OSF deposit and the programme-wide correction record in the corrections log.

v3.5 (7 Nov 2026): the form is set in the programme's LaTeX presentation, still on two printed pages, with its notes, its history and its records after them. The title becomes *The ARC Theory: does control survive recursive self-improvement?*, its established name followed by its question; until this version it read *Two-Page Form · The ARC Theory · Does Control Survive Recursive Self-Improvement?*, and the band above it, *The ARC Theory · The Two-Page Form · Supporting Paper of the Statement Paper*, gives way to the category label and the programme line. The dateline, which until this version began with the author's name, now stands under the author block and ends "from page 3; the version history is Appendix A", where until this version it ended "from page 3 in print"; the line under the title now follows the epigraph and reads "The Statement Paper remains the citable work for the claims.", where until this version it read "Every claim is drawn from the Statement Paper, which remains the citable work for the claims." Law I's status reads "Form heuristic; earlier error-decay fits do not establish its capability exponent.", where until this version it read "Form heuristic; measurement live."; Law II's reads "Proved for positive gain drift in the minimal model (Paper X), for relative drift.", where until this version it read "Proved in the minimal model, for relative drift." In the paragraph on the dates, the record of 8 December 2024 is a DKIM-signed document, where until this version it was DKIM-sealed. In the paragraph on why now, two of the field's impossibility results, preprints not yet peer-reviewed, argue that safety cannot in general be guaranteed, and this theory takes a different object, where until this version the field's impossibility results proved that worst-case control fails and this theory started where they stop; Hinton's remarks are given as the two reports give them, where until this version he conceded that control ends and that engineered care is the only hope, while offering no mechanism; and the domain is stated in two tiers, each law under test, measured first in artificial minds and never assumed beyond them, where until this version the mathematics of form reached everything composed, stars included, and the theory was set beside physics' theory of everything. The register paragraph says that the cosmological wing is set out in the HRIH monograph, where until this version it read "HRIH, the cosmological wing", and the table's row for ARC-4 reads as the registration registers the rule and the Statement Paper prints it, where until this version it read "ARC-4 falls and takes ARC-5 with it". The openness sentence says that a laboratory may run the decisive trial with its own code, since the programme's code licence grants reading only, and the invitation no longer says that the entry point costs nothing. The foot names the Statement Paper by its current title, *The ARC Theory: Growth, correction and alignment in recursive systems*, at version 7.7, where until this version it read *The ARC Theory · Five Laws of Recursive Growth, Correction and Persistence*, and no longer repeats the author's name, ORCID and licence, which the first page and the licence line carry; the version history becomes Appendix A, and the author's disclosure stands under Acknowledgements. Note markers are linked, the first-page footnote gives the theory's placement and two links about the author, the references list the programme's own works the form names, and a priority record and a notation section follow this history; Law II's dated record is given there. Substantive: the status lines of Laws I and II, the description of the record of 8 December 2024, the paragraph on why now, the table's row for ARC-4 and the openness sentence. Editorial: the rest. No statement of a law, kill condition, threshold or registered proposition changes; the one dependency restated, ARC-4's, is printed as registered; no result is added, no date of any claim moves, and the first-published date does not move.

v3.4 (31 Oct 2026): the status is brought to 31 October 2026, and the result of the programme's first registered study to reach a verdict is added. The line on the laws now says that each is a law under test, where until this version it said that neither of Laws IV and V had yet been tested and cited their mathematics at version 1.1 of 7 October 2026; it ends with the sentence on Law IV's two limbs, which until this version followed Law V and named version 1.103, and now names version 1.104, the registration's

current version. Law IV's entry ends with its status, not yet tested, and note G cites the mathematics of Laws IV and V at version 1.2 of 29 October 2026. The sentence after the table, which said that Unit 83 awaited registration and that the theory had not yet been at risk, now gives Unit 83's result on P21 with the DOI of its results. Law II's line takes the wording P4 of the registration gives it, where until this version it read "Self-improvement holds together only while correction out-scales drift." A seventh dated note, G, records where the laws and the trials stand. Substantive: Law II's wording at the boundary, and the first registered result. Editorial: the dateline and the structured and citation records carry the new version and date, and the citation file names version 1.104. No kill condition, dependency, threshold or registered proposition changes, nor the grade of any of Laws I to III; no date of any claim moves, and the first-published date does not move.

v3.3 (18 Oct 2026): the form is set on two printed pages, as its title says; the printed copy of version 3.2 ran to five. The five dated notes, the foot and this history follow it from page 3, each note marked where it applies and kept in its own words. One status line now serves the five laws: each is a law under test until a registered test returns its verdict, and Laws IV and V were registered as laws on 15 October 2026 in version 1.103 of the registration, untested, their mathematics cited at version 1.1 of 7 October 2026. Law V's build-order limb takes the direction version 1.103 registers, where until this version it read $|\gamma_{\text{first}} - \gamma_{\text{after}}| \geq 0.10$, as the statement paper prints it; a line after Law V names Law IV's second limb; the sentence on the decisive trials says where they stand; and a sixth dated note, F, records what version 1.103 registers, how it narrows the rows for ARC-2 and ARC-4, and the state of the first registered study. The entry for v2.1 names four dated notes; three were printed, and the fourth, on what the seal of 8 December 2024 establishes, was held for a decision and is not in this form. Substantive: the direction of Law V's build-order limb. Editorial: the line under the title no longer says the laws are in two pages and, with the band under the dateline folded into it, says once that every claim, not every sentence, is drawn from the statement paper; the dateline states the licence, CC BY 4.0, and where the notes and history are; the foot no longer repeats this note's DOI or the web address of the statement paper; the dateline and the structured and citation records carry the new version and date; note C and the entry for v3.0 link the mathematics by its DOI, their words unchanged. No other statement of a law, and no grade, kill condition, dependency, threshold or registered proposition, changes; no result is added, no date of any claim moves, and the first-published date does not move.

v3.2 (15 Oct 2026): the printed copy no longer carries the website's closing panel on its last page; the words are unchanged.

v3.1 (5 Oct 2026): the title now follows the programme's uniform order (paper, its law or framework, its own title), *Two-Page Form · The ARC Theory · Does Control Survive Recursive Self-Improvement?*; until this version its heading read *Does Control Survive Recursive Self-Improvement?* and its title *Does Control Survive Recursive Self-Improvement? The ARC Theory, in Two Pages*. Mentions of the programme's papers in this paper now carry their uniform titles. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.0 (3 Oct 2026): the laws section states five laws, following version 7.1 of the statement paper. Laws IV and V, added to the theory on 24 September 2026, are named with their equations, Law IV's $S_{\text{ext}}(1) < 1/2$ and Law V's $|\gamma_{\text{first}} - \gamma_{\text{after}}| \geq 0.10$ with a placement gap that widens as depth rises, and with their registers and their status: neither is registered as a law or tested, their mathematics is set out at version 1.0 of 3 October 2026, and the propositions that carry them were registered on 8 September 2026. Two dated notes say what the registration does and does not register and how the rows for ARC-4 and ARC-5 stand beside the registered refuters. Substantive: the count of laws. Editorial: the dateline and the page's structured and citation records carry the new version and date. No sentence on Laws I to III changes, and no kill condition, date, result, threshold or registered proposition changes; the first-published date does not move.

v2.1 (21 Sep 2026): four dated notes are added after an owner review of 19 September 2026, and no

sentence of the paper's body is altered. They state what four compressed passages do and do not establish: that write access makes a reachable constraint alterable without by itself showing that it is ever altered, and that a stronger claim owes its threat model and its behavioural and operational assumptions; that Law II compares growth rates carrying their own coefficients rather than levels, and that the margin is zero at equality, where the coefficients and the specified dynamics decide the outcome; that a domain signature, once verified, establishes the integrity of the signed bytes and asserts a carriage time on the signing domain's authority, which is an assertion and not an independent attestation; and which designs the openness sentence is about, read with this page's own deferral of the licence and with a state owed for each instrument. Substantive: each note bounds a claim the two-page compression had left wider than the record supports. Editorial: the dateline and the page's structured and citation records carry the new version and date, and this version adds the version history block, which the paper did not carry before. No claim, date, result, law, threshold or registered proposition changes, and the first-published date does not move.

B Priority record

Table 1 gives each dated record bearing on this form, with its kind of timestamp, and Appendix B.2 sets out Law II's dated record.

B.1 Dated records

Table 1. Dated records bearing on this form, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
8 December 2024	The claim, written whole in the author's manuscripts of *Infinite Architects*	Self-emailed message of 8 December 2024; manuscripts on OSF QASRK from 5 October 2026, its message headers from 15 October 2026	Time signed by the mail servers (DKIM, 02:45:30 UTC; Google's Authenticated Received Chain (ARC) seal, 02:45:31 UTC), headers public; the deposits dated by OSF's server
2 January 2026	*Infinite Architects* published, first edition	ISBN 978-1806056200; the first edition as printed, on OSF dqm3e since 26 August 2026	Publication date of the first edition, the print run's date on the author's record; its imprint line reads "First Published Tuesday 6th January 2026", which the author's deposit of the first edition on OSF records as a production error
17 January 2026	Law I, the ARC Principle, in Paper I: $U = I \times R^{\alpha}$, capability as a power of recursive depth, with an exponent to be measured (named Law I on 15 August 2026)	OSF file revision 1, node 6c5xb, 09:18:20 UTC, PDF page 2	Deposit dated by its server
22 January 2026	Paper II, version 11.0: a power-law alignment term beside a constant external constraint, an earlier form of Law II's comparison (Appendix B.2)	OSF file revision 1, node 6c5xb, 23:43:26 UTC	Deposit dated by its server; each page also prints "Paper II v11.0 | 22/01/2026" (a date written inside the document)
9 February 2026	Paper III, version 6.1: $S \propto R^{-\alpha}$ (external constraint) and $S \propto R^{0}$ (embedded values)	OSF file revision 1, node hqcgf, 20:06:08 UTC	Deposit dated by its server; the document prints "Version 6.1 | 9 February 2026"

Date	Record	Identifier	Kind of timestamp
20 and 22 February 2026	Paper III, version 9.0: $S \propto R^{\alpha_{\text{align}} - \alpha_{\text{cap}}}$	OSF file revision 7, node hqcgf, 22 February 2026, 08:15:13 UTC	The document prints "Version 9.0 | 20 February 2026"; its deposit is dated by OSF's server on 22 February 2026
26 June 2026	The correction-and-drift formulation of Paper X: $\beta > k$, the equilibrium relation and the hard-takeoff result	GitHub pull request 1, "Paper X: The Coupled Co-Scaling Law (v4.1)", created 12:57:55 UTC, merged 16:25:53 UTC	Record dated by GitHub's server; the repository's visibility at that time is not reconstructed
14 August 2026	The ARC Theory, the Theory of Artificial Recursive Creation, in the Statement Paper	OSF file revision 1, node gw5mx, 07:33:39 UTC	Deposit dated by its server
16 August 2026	Law III, the ARC Ceiling: $\alpha_{\text{crit}} = 1/(1-\gamma)$, the growth exponent capped by the reciprocal of the correction shortfall, in the Statement Paper; an earlier form, $1/\gamma$, printed from 14 August 2026, was corrected on 16 August 2026 (named the ARC Ceiling on 17 August 2026)	OSF file revision 27, node gw5mx, 04:06:08 UTC	Deposit dated by its server
17 August 2026	This form first published: its HTML, PDF and citation file	OSF node mc5xw, files created from 04:53:39 UTC	Deposit dated by its server
8 September 2026	The ARC Theory registration, version 1.100: twenty-two propositions, and Laws I to III (updated 13 September 2026, version 1.102; 15 October 2026, version 1.103; and 29 October 2026, version 1.104, the current version)	DOI 10.17605/OSF.IO/P8CKQ	Registration accepted on OSF
29 September 2026	Laws IV and V, the ARC Persistence Law (register ARC-4, carrying P15) and the ARC Embedding Law (register ARC-5, carrying P12 and P21), stated in the treaty paper on the HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence (added to the theory on 24 September 2026)	OSF file revision 1, node wxpce, 09:36:17 UTC, PDF pages 7 and 9; their mathematics first deposited on OSF 4b3vt on 3 October 2026	Deposit dated by its server; a date written inside the document
15 October 2026	Laws IV and V registered as laws by version 1.103 of the registration, "the text that registers them as laws"	DOI 10.17605/OSF.IO/P8CKQ, version 1.103	Registration update approved on OSF

The programme's Priority Index (version 2 of 26 October 2026, a working draft) gives the earliest dated record of each of the author's major claims, with its quotations, addresses and concessions to earlier work.

B.2 Law II's dated record

Law II's dated record has two parts, kept apart: the earliest statement of its idea, and the earliest dated record of its inequality, $\beta > k$.

The idea. The conceptual thesis, that recursion amplifies, and that stable recursion requires correction that co-scales with the amplification, was first set out in the author's book manuscript *Infinite Architects* (in a message the author sent himself on 8 December 2024, dated by the signed header time of the mail servers that carried it, now on OSF; published 2 January 2026); the formal results of Paper X are the 2026 measurable form of that thesis, and are *not* claimed to appear in the book. The comparison Law II sharpens was stated in words, as pace, in the print manuscript of *Infinite Architects* of 30 April 2025, page 135: "no line of external code or naive policy can keep pace" (in a message the author sent himself, its time signed by the mail servers and the message held privately; the manuscript on OSF from 5 October 2026). The co-scaling intuition is not claimed as new: Paper X credits the law of requisite variety of Ashby (1956), the good-regulator theorem of Conant and Ashby (1970) and the scaling laws for scalable oversight of Engels et al. (2025), which give oversight success as a function of a capability gap and recommend games in which the overseer's performance rises more steeply with general capability than the adversary's, two slopes in a contest between models rather than a condition on a self-improving system's correction against its own drift; Paper X claims the explicit criterion. Kornai et al. (2023) ask, qualitatively, for mechanisms that strengthen as an agent evolves, with no rate or condition.

Paper II, version 11.0 of 22 January 2026, set a power-law alignment term beside a constant external constraint (dated PDF) and stated the comparison in words: "External constraints are eventually overwhelmed" . . . "Only alignment embedded in reasoning can maintain pace with capability." That is an earlier form of Law II's comparison: alignment imposed from outside does not keep pace with recursive capability, and only alignment embedded in reasoning does. It was argued, and partly measured in one laboratory; the registration records the reading as contestable. Paper III, version 6.1 of 9 February 2026, expressed the alignment-to-capability ratio as $S \propto R^{-\alpha}$ for its external-constraint case and $S \propto R^{0}$ for its embedded case (dated PDF). The Foundational Paper, version 1.1 of 11 February 2026, wrote the ratio as $S \propto R^{\alpha_{\text{align}}-\alpha}$ (dated PDF), and Paper III, version 8.2 of 13 February 2026, wrote the comparison as a general inequality, if $\alpha_{\text{cap}} > \alpha_{\text{align}}$ then $S \propto R^{\alpha_{\text{align}}-\alpha_{\text{cap}}} \to 0$ (dated PDF); version 9.0, dated 20 February 2026 and deposited on 22 February 2026, kept it. A related comparison on another axis came three days after Paper II and before these general forms: Engidaw (2026) (Zenodo, 25 January 2026) states that if capability grows with training compute faster than alignment overhead, a safety margin defined as overhead over risk-weighted capability falls below one past a finite compute threshold. It has no recursive depth, no misalignment fraction and no drift arising from improvement; read with its overhead as Paper X's correction strength, its condition is $\beta > 1$, which coincides with $\beta > k$ only at $k = 1$. The direction in qualitative form (an evolving agent outgrows constraints imposed from outside) is earlier in Kornai et al. (2023), without recursive depth, an exponent or any measurement.

The inequality, $\beta > k$. The inequality's earliest dated record in the programme is the correction-and-drift formulation of Paper X, of 26 June 2026, whose priority disclosure in version 4.0 states "(ii) the sharpened stability criterion" (a commit date). The repository supplies a server-dated record for this formulation: pull request 1 was created on 26 June 2026 at 12:57:55 UTC and merged that day at 16:25:53 UTC. Its description states the correction-strength criterion, equilibrium relation and hard-takeoff result. This is distinct from a date written inside a manuscript and from a committer-supplied date. The current public record does not, by itself, reconstruct every past repository-visibility setting. OSF deposits followed on 2 July 2026 (the README) and 4 July 2026 (the paper).

This date records the later correction-and-drift formulation developed in Paper X; it is not the first date of the programme's co-scaling equations. The earlier depth-scaling exponents and Paper X's correction-and-drift exponents belong to their respective stated models. The earlier papers do not establish every theorem developed there. Earlier and later parameters retain their respective meanings: the February self-referential-coupling parameter is not automatically Paper X's correction-strength exponent.

Similar algebraic form does not identify different model quantities. The earlier depth-scaling comparison establishes a mathematical precursor; the later relative-drift dynamics and their theorem must be dated from their own records. The inequality was named Law II on 15 August 2026, and registered as β_C exceeds k on 8 September 2026 (registration).

Later work: Canale (2026) (arXiv:2610.08540, 6 October 2026) models the alignment burden of each risk as a power of capability and compares its exponent with that of a budget proportional to capability, with the equal case decided by coefficients; its correction capacity is fixed at the first power of capability and its models do not improve themselves, so it neither tests nor contradicts the criterion, whose three regimes the theory registration fixed on 8 September 2026.

C Notation

Each symbol and label the form uses is given here in the words of the form, of its notes or of the registration (version 1.104).

U, I and R. Usable capability, intelligence and recursive depth, in Law I, $U = I \times R^{\alpha}$. In the form's words, "Intelligence converts recursion into capability"; in the registration's, "usable capability is a power of recursive depth".

α. The capability-growth exponent of Law I, "measured rather than assumed" (the registration). An error-suppression exponent is a different quantity from the capability exponent: no capability growth is derived from an error-decay fit unless a metric bridge is stated and identified.

β and k. The exponents Law II compares, $\beta > k$: β, which the registration writes β_C, is the correction-strength exponent, and k is the drift exponent, so that "correction has the asymptotic scaling advantage where its exponent exceeds the drift exponent". In the registration's words, β_C "is not β_L, the leverage fraction of the recursive step that appears in the coupling theorem".

γ. The correction-leverage exponent, which "measures how well a system's error-correction keeps pace as the system grows" (the form's Law III).

α_{crit}. The ceiling of Law III, the ARC Ceiling, $\alpha_{\text{crit}} = 1/(1-\gamma)$: "the growth exponent is capped by the reciprocal of the correction shortfall" (the registration), $1-\gamma$ being the corrector's shortfall from keeping perfect pace.

The ARC Bound. $\alpha \leq 2$, the ceiling's value two, which α_{crit} takes at $\gamma = 1/2$. The form states it as the conjectured value, "a stability limit rather than an impossibility limit"; the registration names the law, the ARC Ceiling, and the value, the ARC Bound, apart.

$S(x)$, $Y(x)$, x and $S_{\text{ext}}(1)$. Law IV's measure: $S(x) = Y(x)/Y(0)$ is "the gain remaining at budget ratio x over the gain installed" (the form). $S_{\text{ext}}(1)$ is that fraction for the externally installed gain at $x = 1$, where the capability-training budget matches the alignment budget that installed it. Law IV's retention limb, P15's claim, is $S_{\text{ext}}(1) < 1/2$: "more than half of what was installed is lost" at that budget.

γ_{first} and γ_{after}. The correction-leverage exponents of "a corrector formed before the capability it corrects" and of "one added after it, at matched content" (note F). Law V's build-order limb, read on P12, is $\gamma_{\text{first}} - \gamma_{\text{after}} \geq 0.10$, the registered minimum effect on that exponent's scale.

***R* in P21.** Recursive depth in Law V's placement limb, along which "the gap between correction inside the loop and correction applied to the finished output widens as depth R rises" (the form).

Δ. The balance exponent, "the correction service's growth elasticity minus the burden's": on one observed trajectory, the measurable quantity, "estimable from logged series a laboratory already has, without adopting any model in this theory" (the form).

ARC-1 to ARC-5. The theory's five independently falsifiable hypothesis registers, one for each law, with the propositions each holds (the registration): ARC-1, recursive capability scaling (Law I: P1, P2, P8, P9 and P18); ARC-2, correction co-scaling (Law II: P4 alone); ARC-3, the ceiling (Law III: P3, P6, P16, P17, P20 and P22); ARC-4, value persistence (Law IV: P15 alone); and ARC-5, the genesis strategy (Law V: P12 and P21).

P1 to P22. The twenty-two propositions of the registration, registered on 8 September 2026. Seven belong to no register: P5, P7, P10, P11, P13, P14 and P19.

Unit 06. The corrector-class audit, registered on OSF on 1 October 2026 (DOI 10.17605/OSF.IO/RV2B5) and run on 6 October 2026 (results: DOI 10.17605/OSF.IO/4BPHG); it tests no law.

Unit 82. The counted test of the premise beneath Law III's value two, P17: whether correctors drawn from one engine miss the same planted departures more often than independence allows. Registered on OSF on 27 October 2026 (DOI 10.17605/OSF.IO/JUFG4).

Unit 83. The counted test of P21 under Law V, registered on OSF on 22 October 2026 and public from 25 October 2026 (DOI 10.17605/OSF.IO/MN73W), and run on 27 and 28 October 2026 (results: DOI 10.17605/OSF.IO/98HKR).